Solutions and R Programming Examples: Confidence Interval

1.Fast Food

1.1) 7.093 1.2) 0.363 1.3) 0.779

1.4) (6.315,7.872)

We are 95% confident that the true population mean of the amount spent for lunch at a fast-food restaurant is between $6.315 and $7.872.

R Program

```
food<-c(7.42,6.29, 5.83, 6.50,8.34, 9.51,7.10, 6.80,5.90, 4.89,6.50, 5.52, 7.90, 8.30, 9.60)

m<- mean(food)

m

[1] 7.093333

sd<-sd(food)

n<-15

se<- sd/sqrt(n)

se

[1] 0.3630357

moferr<-qt(0.975,n-1)*se

moferr

[1] 0.7786341

 lcl<-m-moferr

 ucl<-m+moferr

 lcl

[1] 6.314699
```

ucl

[1] 7.871967

………

2. The complaints

2.1) 43.040 2.2) (31.125, 54.955)

………

3. Flu shot

3.1) 0.00615 3.2 0.00125 3.3) 0.0025 3.4) (0.0037, 0.00861) 3.5) Yes

………

4. Basic life

4.1) 0.54255, 0.61381 4.2) (-0.12879, -0.0137)

R Program

```
prop.test(x=c(306,391), n=c(564,637),conf.level=0.95)
```

………

```
        2-sample test for equality of proportions with continuity correction

data:  c(306, 391) out of c(564, 637)
X-squared = 5.9483, df = 1, p-value = 0.01473
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.12878962 -0.01373351
sample estimates:
   prop 1    prop 2
0.5425532 0.6138148
```

5. Study time

5.1) 2.567 **5.2) (-3.110, 8.244)** We are 95% confident that the true population mean of study time per week between the two groups (students who planned to go to graduate school and those who did not) is between -3.110 – 8.244 hours.

This means that there is no statistically significant difference in the true mean of study time per week between the two groups.

R Program

```
plan<-c(15,7,15,10,5,5,2,3,12,16,15,37,8,14,10,18,3,25,15,5,5)

N_plan<-c(6,8,15,6,5,14,10,10,12,5)

t.test (plan,N_plan,conf.level=0.95,var.equal=TRUE)
```

....

Two Sample t-test

data: plan and N_plan

t = 0.92469, df = 29, p-value = 0.3628

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-3.110283 8.243616

sample estimates:

mean of x mean of y

11.66667 9.10000

..

6. Multiple myeloma

6.1) 86.143 **6.2) (-28.261, 200.547)** We are 95% confident that the true population mean difference in bone marrow microvessel density of patients who received transplants of stem cells (before and after) is between -28.261, 200.547.

This means that there is no statistically significant difference in the true mean of bone marrow microvessel density between the two groups.